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Google Scholar Profile

Himanshu Kumar, Ph.D.

POST-DOCTORAL FELLOW IN NEUROENGINEERING & EPILEPSY

RESEARCH INTERESTS

Affective Computing, Cognitive Neuroscience, Epilepsy, EEG and Stereotactic EEG Analysis, Biomedical Signal Processing, Machine Learning, Explainable AI (XAI) and Deep Learning.

WORK EXPERIENCE

Post-Doctoral Fellow — **Oct 2024 – Present | Cleveland, OH, USA**

Neurological Institute, Cleveland Clinic

Stereotactic EEG (sEEG) signal processing, epilepsy network modeling, seizure segmentation, and multivariate changepoint analysis.

Project Officer — **Jan 2024 – Aug 2024 | Chennai, India**

Department of Biomedical Engineering, Indian Institute of Technology (IIT) Madras

Worked on Indian Medical Device Regulations and Standards. Investigated multimodal biosignal integration (EEG/EDA/PPG).

Guest Scientist — **Mar 2021 – Sep 2022 | Aachen, Germany**

University Hospital RWTH Aachen (Dept. of Psychology, Psychotherapy and Psychosomatics)

Investigated trait impulsivity network classification using EEG connectome predictive modeling and Explainable Machine Learning (XAI).

EDUCATION & TRAINING

Doctor of Philosophy (Ph.D.) in Biomedical Engineering — **2019 – 2024 | Chennai, India**

Indian Institute of Technology (IIT) Madras

Advisors: Prof. S. Ramakrishnan, Dr. Subha D. Puthankattil

Thesis: *"Analysis and classification of audio-visual stimuli induced arousal and valence emotional states using EEG signal"*

Integrated Master of Technology (M.Tech.) in Biomedical Engineering — **2013 – 2018 | Prayagraj, India**

Indian Institute of Information Technology (IIIT) Allahabad

Five-year dual degree (B.Tech. & M.Tech.). CGPA: 9.0/10.0. Awarded Academic Excellence Bronze Medal.

FELLOWSHIPS & GRANTS

DAAD Bi-National Guided Research Fellowship — **2021 – 2022 | Germany**

German Federal Foreign Office / RWTH Aachen University Hospital

Advanced Research Opportunities Program (AROP) — **2021 | Germany**

RWTH Aachen University

Half Time Research Assistantship (HTRA) — **2019 – 2023 | India**

Ministry of Education (formerly MHRD), Government of India (IIT Madras)

Institute Travel Award — **2022 | UK**

IEEE Engineering in Medicine and Biology Conference (EMBC), Glasgow

PUBLICATIONS

INTERNATIONAL JOURNALS

1. **Himanshu K**, Nagarajan G, and Ramakrishnan S. (2025) 'Analysis of dynamics of EEG signals in emotional valence using super-resolution superlet transform' *IEEE Sensors Letters*. **IF: 2.2**
2. Philippa H., **Himanshu K**, Ramakrishnan S, Ute H, Carmen W. (2024) 'Functional Brain Networks of Trait Impulsivity: A Connectome-Based Predictive Modeling Analysis.' *Human Brain Mapping*, 45(15), e70059. **IF: 3.5**
3. Philippa* H., **Himanshu* K**, Aliaksandra S, Ramakrishnan S, and Ute H (2023) 'Impulsivity Classification Using EEG Power and Explainable Machine Learning.' *International Journal of Neural Systems*, 33.02: 2350006 (*Joint First Author). **IF: 8.0**
4. **Himanshu K**, Nagarajan G, Subha D.P, and Ramakrishnan S. (2024) 'Analysis of EEG Fluctuation Patterns using Non-Linear Phase Based Functional Connectivity Measures for Emotion Recognition.' *Fluctuations and Noise Letters*, doi: 10.1142/S0219477524500512. **IF: 1.8**
5. Saurabh B, **Himanshu K**, Nagarajan G, Ramakrishnan S (2024) 'Exploring Central-Peripheral Nervous System Interaction Through Multimodal Biosignals: A systematic Review' *IEEE Access*, doi: 10.1109/ACCESS.2024.3394036. **IF: 3.9**
6. Saurabh B, **Himanshu K**, Nagarajan G, Ramakrishnan S (2024) 'Assessment of Emotion Elicitation using Multimodal Physiological Sensors and Phase Synchronization' *IEEE Sensors Letters*, doi: 10.1109/LSENS.2024.3426562. **IF: 2.2**
7. Yedukondala Rao V, **Himanshu K**, Nagarajan G, Balasubramaniam N, and Ramakrishnan S. (2021) 'A Systematic Review of Sensing and Differentiating Dichotomous Emotional States Using Audio-Visual Stimuli', *IEEE Access*, 9, pp. 124434–124451. **IF: 3.9**
8. Nagarajan G, Yedukondala Rao V, **Himanshu K**, and Ramakrishnan S. (2021) 'Emotion recognition using Electrodermal activity signals and multiscale deep convolutional neural network', *Journal of Medical Systems*, 45(4), pp. 1-10. **IF: 5.3**
9. **Himanshu K**, Nagarajan G, Subha D P, & Ramakrishnan S. (2021). EEG based emotion recognition using entropy features and Bayesian optimized random forest. *Current Directions in Biomedical Engineering*, 7(2), 767-770. **Scopus Indexed**
10. **Himanshu K**, Nagarajan G, Subha D.P, and Ramakrishnan S. (2021) 'Emotion Recognition in EEG Signals Using Decision Fusion Based Electrode Selection', *Studies in Health Technology and Informatics*, 281: 153-157. **Scopus Indexed**

UNDER REVIEW & REVISION SUBMITTED

11. **Himanshu K**, Satyavratan G, Manuskandan S.R, Karthick P.A, Nagarajan G. (2025) 'EEG-Based Emotion Recognition using Super-Resolution Superlet Transform and self-attention Convolutional Neural Network', *Biomedical Signal Processing and Control* (Revision Submitted). **IF: 5.0**
12. **Himanshu K**, Nagarajan G., Subha D.P, Ramakrishnan S. (2025), 'Subject-Independent Emotion Recognition with EEG Bispectral Quadratic Phase Coupling Features and Explainable Machine Learning', *Signal, Image and Video Processing* (Revision Submitted). **IF: 2.2**
13. **Himanshu K**, Guhan N.P, David M, Imad N, Andreas A, Juan C.B, Demitre S, Balu K, (2026), 'Three-Phase Seizure Segmentation in Stereotactic EEG Using Envelope-Based Multivariate Changepoint Analysis', *Annals of Biomedical Engineering* (Under Review).

SELECTED CONFERENCE PROCEEDINGS

1. Sourabh B, **Himanshu K**, Nagarajan G, Ramakrishnan S. (2024) 'Assessment of EEG-PPG Cross Frequency Coherence under Evoked Emotional Arousal' *BMT 2024*.
2. Sourabh B, **Himanshu K**, Nagarajan G, Ramakrishnan S. (2024) 'Assessment of Valence Emotional State Using EEG-EDA Coupling and Explainable Classifiers' *MIE 2024*.
3. **Himanshu K**, Nagarajan G, Subha D.P, and Ramakrishnan S. (2022) 'Assessment of emotional states in EEG signals using multi-frequency power spectrum and functional connectivity patterns' *IEEE EMBC 2022*.
4. **Himanshu K**, Philippa H, Ramakrishnan S, Ute H (2022) 'Impulsivity classification using EEG-based features' *OHBM 2022*.
5. **Himanshu K**, and Ramakrishnan S. (2022) 'Time and Frequency domain analysis of APB muscles Abduction in adult dominant hand using surface electromyography signals' *IEEE MeMeA 2022*.
6. **Himanshu K**, Nagarajan G, Subha D.P, and Ramakrishnan S. (2022) 'Classification of Emotional States Using EEG Signals and Wavelet Packet Transform Features' *MIE 2022*.
7. **Himanshu K**, Nagarajan G, Subha D.P, and Ramakrishnan S. (2022) 'Assessment of Emotional States in EEG Signals Using Phase Slope Index Based Functional Connectivity Features' *RMBS 2022*.

INVITED LECTURES

Preconference Lecture at 4th IEEE Sensors and Related Networks Conference — Jul 2025 | VIT, Vellore

Next Generation Health Technologies: Signal Processing, Teleneurology & AI — Jun 2025 | AISAT, Kerala

Medical Informatics and Digital Health Programme — Mar 2025 | IIT Hyderabad

BME Engineering Training Program (IITM & ISRO Human Spaceflight Training) — Feb 2025 | IIT Madras

PERSONAL SKILLS

PROGRAMMING & FRAMEWORKS

Python, MATLAB, C, Java, PyTorch, TensorFlow, Keras, Scikit-learn

SOFTWARE & ANALYSIS TOOLS

EEG Lab, BrainVision Analyzer, FreeSurfer, SPM12, LabView, Origin Pro

HARDWARE & RESEARCH PLATFORMS

fMRI Acquisition, BITalino (Biosignals), NI USB 6363, Biopac MP36, Raspberry Pi

OTHER DETAILS

Date of Birth: 24/09/1994 | Nationality: Indian | Gender: Male